THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** $2^4 \times 3^3$	16×27
2	**(c)** 12	$2^2 \times 3$
3	**(d)** $a = c$	Product $\frac{c}{a} = 1$
4	**(b)** ± 1	$(x-1)(x+1)$
5	**(d)** no solution	$\frac{3}{6} = \frac{4}{8} \neq \frac{5}{7}$
6	**(b)** 3	$21a = 63$
7	**(c)** $x^2 + 2x - 63 = 0$	$x(x+2) = 63$
8	**(a)** 32	$8 + 24$
9	**(a)** Student A is correct	$12, \ldots, 248$: $\frac{236}{4} + 1 = 60$
10	**(b)** 21	$24 - 3$
11	**(c)** Incorrect — they are similar, not necessarily congruent	AA gives similarity
12	**(b)** Incorrect — equal area does not imply equal angles	Area alone decides nothing
13	**(c)** 1	$\sin^2 A + \cos^2 A$
14	**(c)** Incorrect — it is perpendicular to the radius through that point	Theorem 10.1
15	**(a)** 154 cm²	$\frac{1}{4} \times 616$
16	**(d)** 308 cm²	Semicircle
17	**(a)** 718.67 cm³	$\frac{2}{3} \times \frac{22}{7} \times 343$
18	**(c)** 10	$20 - 10$
19	**(b)** Incorrect — both give the same mean	Only arithmetic is simplified
20	**(d)** $\frac{1}{4}$	Only TT

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$\frac{306 \times 657}{22338} = 9$	2
22	$3(x-5)(x+5) \Rightarrow \pm 5$; sum $0 = -\frac{0}{3}$, product $-25 = \frac{-75}{3}$ ✓	1+1
23	$D = 9 - 40 = -31 < 0 \Rightarrow$ no real roots	1+1
24	$EC = 12$ cm $\Rightarrow AC = 20$ cm	1+1
25	$B = (10-2, -2-3) = (8, -5)$	1+1
26	$\frac{1/\sqrt{2}}{\frac{2}{\sqrt{3}}+2} = \frac{3\sqrt{2}-\sqrt{6}}{8}$	1+1
27	$10 \tan 60^\circ = 10\sqrt{3} \approx 17.32$ m	1+1
28	$2(4+3+5) = 24$ cm	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,-6), (3,0)$; $(0,0), (1,-1)$. Intersect at $(2,-2) \Rightarrow$ consistent	2+1
30	cf 5, 14, 28 $\Rightarrow 30 + \frac{20-14}{14} \times 10 = 34.29$	1+1+1
31	$4d = 16 \Rightarrow d = 4$; $a + 28 = 31 \Rightarrow a = 3 \Rightarrow 3, 7, 11, 15, \ldots$	1+1+1
32	Let M be the mid-point of arc AB. OM bisects $\angle AOB$, so $OM \perp AB$. Also $OM \perp$ tangent at M $\Rightarrow$ tangent $\parallel AB$	1+1+1
33	Pond $= 231$ m²; remaining $= 1386 - 231 = 1155$ m²	2+1
34	$\frac{7}{12}$; $\frac{6}{11}$	2+1

35	$30\sqrt{3} - \frac{30}{\sqrt{3}} = 20\sqrt{3} \approx 34.64$ m	1+1+1
	OR: $\tan\theta = \frac{28.5}{28.5} = 1 \Rightarrow 45°$; for $60°$: $\frac{28.5}{\sqrt{3}} \approx 16.45$ m	1+1+1
36	$\frac{n}{2} = 200$; cf 14, 70, 130, 216 ⇒ $3000 + \frac{70}{86} \times 500 = 3406.98$ hrs	1+1+1
	OR: $a = 70,\ h = 10$; $\Sigma fu = -6 - 10 + 0 + 8 + 6 = -2$; mean $= 70 - \frac{20}{35} = 69.43\%$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	$AB^2 = BC^2 = 106$, $AC^2 = 212 = AB^2 + BC^2$ (3). Mid-point of AC $= (5, 3)$: distances to A, B, C all $\sqrt{53}$ (2)	3+2
	OR: $\frac{24a+3a+1}{4} = 9a - 2 \Rightarrow a = 1$; $\frac{15-3}{4} = 3 = -b \Rightarrow b = -3$; A(4, −3), B(8, 5): $AB = 4\sqrt{5}$	3+2
38	$= \frac{\sin A(1-2\sin^2 A)}{\cos A(2\cos^2 A - 1)}$; $1 - 2\sin^2 A = 2\cos^2 A - 1 \Rightarrow \tan A$ (3). $A = 30°$: $\frac{\frac{1}{2} - \frac{1}{4}}{\frac{3\sqrt{3}}{4} - \frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$ ✓ (2)	3+2
	OR: RHS $= \frac{(1-\sin\theta)^2}{\cos^2\theta} = \frac{(1-\sin\theta)^2}{(1-\sin\theta)(1+\sin\theta)}$	2+2+1
39	(i) $\frac{2}{3}\pi r^2 h = 2464$ cm³ (ii) $l = 25$: $1056 + 154 + 550 = 1760$ cm²	3+2
	OR: (i) $0.5\pi + \frac{\pi}{6} = \frac{2\pi}{3} \approx 2.10$ m³ (ii) $2\pi + \pi = 3\pi \approx 9.43$ m²	3+2
40	Mean $= \frac{2500}{50} = 50$ (2); median: cf 7, 17 ⇒ $40 + \frac{25-17}{15} \times 20 = 50.67$ (3)	2+3
	OR: (i) $\frac{1920}{60} = 32$ (ii) cf 4, 12, 24 ⇒ $30 + \frac{30-24}{20} \times 10 = 33$ (iii) $3(33) - 2(32) = 35$	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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